

2.1 Sensitivity (the derivative)

Study Notes · Third Eye Labs
Season 2, episode 1

THE ONE QUESTION

"If I nudge this number, does the score get better or worse — and by how much?"

Asked once per weight, per step, for the entire history of machine learning.

The setup

one weight w
64 real data points
 $\text{score}(w) = \text{mean}((w \cdot x - y)^2)$
start at $w = 2.0 \rightarrow \text{score } 0.5062$

1. Nudge it and divide

$h = 0.1$

$\text{score}(2.1) - \text{score}(2.0) = -0.02254$

$\div 0.1 = -0.2254$

that ratio is "rise over run" — the thing you already did at school

h	ratio
0.1	-0.22540
0.01	-0.24754
0.001	-0.24975
0.0001	-0.24998
0.00001	-0.25000

The digits stop changing, one at a time. It settles on -0.25 — and that settling number is the derivative. This is a limit actually happening, not a definition.

2. Check it

method	value
shrinking nudge	-0.25000
autograd (PyTorch)	-0.2500000

Agreement to 6 decimals. The number is real, not an artefact of the arithmetic.

4. The symbol, last

dL / dw

⚠ Not a fraction. It is the name of the number you just watched settle. If the notation means nothing to you, ignore it — you already have the idea.

3. What the sign means

- ratio **negative** $\rightarrow$ score falls as w rises $\rightarrow$ increase w
- ratio **positive** $\rightarrow$ score rises as w rises $\rightarrow$ decrease w

The sign is the instruction. Walking downhill from $w = 2.0$ lands at $w = 2.9993$ — and the data was really made with $w = 3.0$.

5. Advanced — where this lives

- Partial derivative: the same question with the other knobs held still. Stack them and you get the gradient $\rightarrow$ 2.2
- Autodiff: frameworks never use shrinking nudges — too slow, too noisy. They apply the chain rule symbolically $\rightarrow$ 2.4
- Finite differences still matter as a gradient check: if your hand-written backward pass disagrees with a nudge, the backward pass is wrong.

- In a transformer: this exact question, asked of 124 million weights, on every single training step.

NEXT: with one weight the answer is a *sign*. With two it has to become a *direction* → 2.2, the *gradient*.